



Leonardo Coscia

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EDUCATION

Master of Science (EQF 7) | Physics (Theory of Quantum Materials)

110 cum laude/110

Sep. 2024 – Jul. 2026

University of Pisa - Pisa, Italy

Bachelor of Science (EQF 6) | Physics

110 cum laude/110

Sep. 2021 – Jul. 2024

University of Pisa - Pisa, Italy

Second Level Master (EQF 8) | Physics

29.8/30 avg.

Oct. 2021 –

Scuola Normale Superiore - Pisa, Italy

RESEARCH EXPERIENCE

PhD candidate | Quantum Technology and Nanoscience

Nov. 2026 –

Scuola Normale Superiore - Pisa, Italy

RESEARCH INTERESTS

My research interests lie in the field of **condensed matter theory**, with a particular focus on:

- many-body physics;
- strongly correlated electrons;
- anyonic phases of matter;
- topological materials.

My Bachelor's thesis explored the mathematical foundations of Chern and $\mathbb{Z}_2$ topological insulators, while my Master's thesis is centered on the investigation of anyonic superconductivity in Moiré materials.

Though I have some experience with computational methods (DFT, DMRG, exact diagonalization, Wannierization), my main interests are **analytical** and **field-theoretical** approaches to condensed matter systems.

PROJECTS AND RESEARCH

Anyon Superconductivity in Moiré Crystals

RPA study of interacting abelian anyons in a periodic potential for the purpose of investigating superconductivity in fractionally filled TMD bilayers

Level Spacing Statistics, Integrability and Many-Body Localization in Quantum Spin Chains

C++, Python - Exact numerical diagonalization (full and partial) of quantum spin chains for the purpose of relating level spacing statistics to integrability and many-body localization

DMRG Study of Quantum Phase Transitions in the Heisenberg XXZ Model

C++, Python - Implementation of the DMRG algorithm in C++ for the purpose of studying phase transitions in the Heisenberg XXZ model

Ab Initio Study of Topological Properties of 2D Crystalline Insulators

Quantum ESPRESSO, Wannier90, WannierTools, Python - DFT calculation, Wannierization and topological analysis of bulk 2D materials, checked against nanoribbon edge states

HONORS AND AWARDS

Admission to Scuola Normale Superiore

Selected through a highly competitive national exam (16 physics students admitted in my cohort) to attend Italy's most prestigious higher education institution with a fully funded scholarship

TEACHING EXPERIENCE

Teaching Assistant at "Stage STEM - Scienziate di Domani"

STEM course for outstanding high school girls

April 2024, May 2025

Scuola Normale Superiore - Pisa, Italy

Lecturer at "Stage di Fisica a Pisa"

Training camp for prospective EuPhO and IPhO competitors

February 2023

Scuola Normale Superiore - Pisa, Italy

Examiner at "Stage di Fisica a Pisa"

Training camp for prospective EuPhO and IPhO competitors

December 2021

Scuola Normale Superiore - Pisa, Italy

Various student teaching projects in High School

Science lessons for elementary and middle school students

Sep. 2015 – Jun. 2021

Liceo Scientifico E. Amaldi - Novi Ligure, Italy

COMMUNITY INVOLVEMENT

Red Cross Volunteer

BLS training, AED certification (now expired), patient transport training (only theoretical)

2019 –

Novi Ligure, Italy

SKILLS

Languages: English (C2), French (B2)

Programming: Python, C++

Numerical Methods: Quantum ESPRESSO, Wannier90, WannierTools

Document Creation: Microsoft Office Suite, LaTeX, Typst